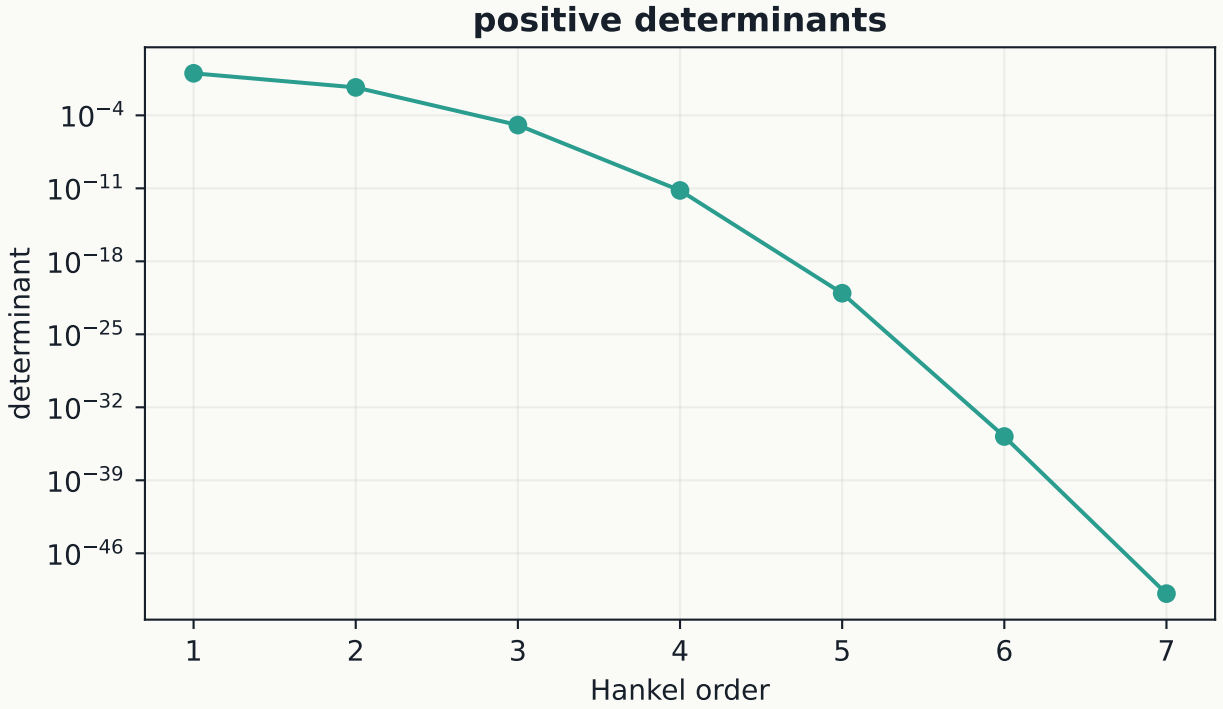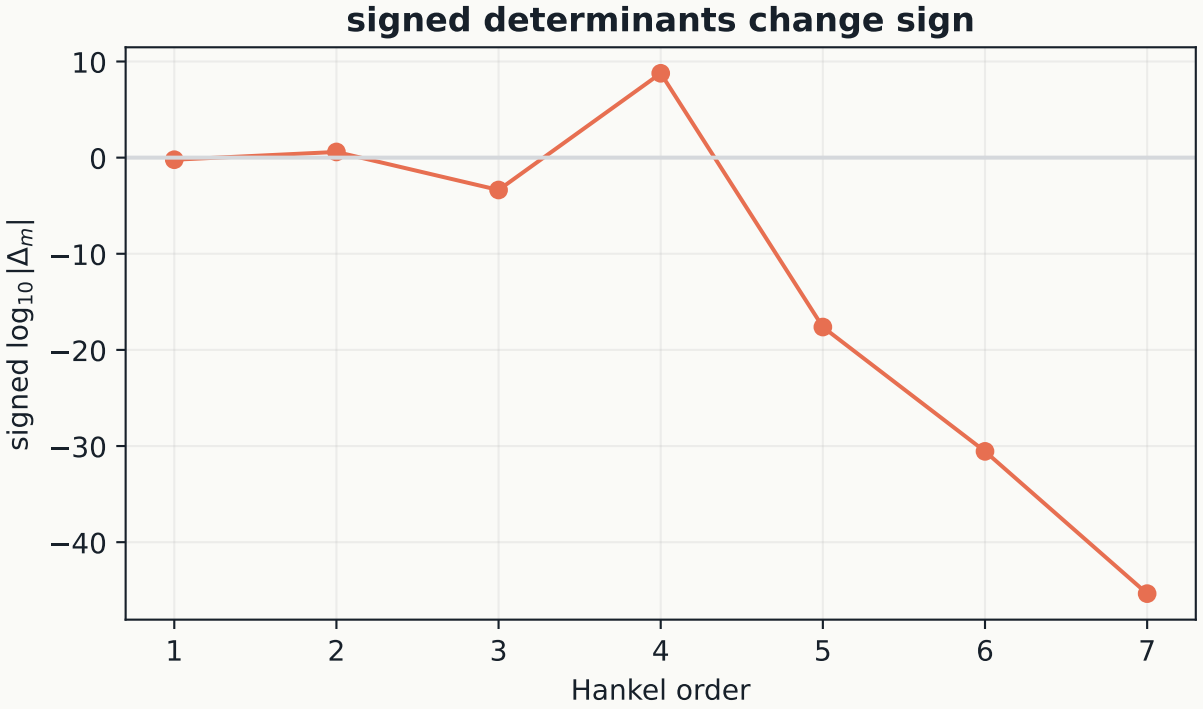The signed moment problem is quasi-definite rather than positive